GridSense AI - Release Note

built 2026-11-06 12:44 UTC - figures fetched from the live deployment and from git at build time

1. Where it stands

Frontend	https://grid-senseai.vercel.app -> HTTP 200
Backend	https://grid-senseai-backend.fastapicloud.dev -> HTTP 200 (health), 200 (validation report)
Model trained on	51,959 hourly rows - 3 zones over 722 days (entsoe 51,959); weather nasa_power 51,689, unavailable 270
API serves from	1,008 in-memory rows - 336 h per zone, the tail kept by ROWS_PER_ZONE_BUFFER = 336
Release window	commits on main dated 2026-11-05 or later
Backend delta	26 commits; 56 files changed, 6903 insertions(+), 653 deletions(-)
Frontend delta	17 commits; 37 files changed, 3862 insertions(+), 782 deletions(-)
Overall verdict	WARN across 13 release gates: 9 PASS, 4 WARN, 0 FAIL

The product is deployed and answers with real data. What follows is deliberately split into what shipped, what broke and was repaired, what is still open, and what the next phase needs - with the per-feed currency table first, because on a grid dashboard 'current' has to mean something measurable.

2. Current condition, per feed

Feed	Source	Rows (served / trained)	Latest hour	Age now	What that means
Load	ENTSO-E transparent day-ahead	1,008 / 51,959	2026-11-04T17:00	43.7 h	real measured hours, but a frozen snapshot: the API serves the committed frame and advances it by rebuild plus redeploy, so this ages with the load
Weather	NASA POWER hourly	738 / 51,689	2026-10-31T23:00	133.7 h	lags load by 3.8 days; 270 served rows have no weather at all, so every live forecast runs on the fallback
Capacity	p99-of-load proxy	derived per zone	n/a	n/a	a distribution peak observed in the data, not an operator's rated capacity
Serving window	backend scheduler	3 zones	2026-11-04T17:00	43.7 h	ok

Served rows count the in-memory tail the API answers from; trained rows count the full feature table the models were fitted on. They are different scopes, not different measurements of the same thing.

Which model is answering right now, per zone:

Zone	Model answering	Horizon	Hour +1 median	Origin age
Spain	no / weather / fallback	24 h	33,219 MW	43.7 h
Greece	no / weather / fallback	24 h	6,202 MW	43.7 h
Italy South	no / weather / fallback	24 h	2,677 MW	43.7 h

All three zones are answered by the reduced-feature fallback because NASA POWER is 90 hours behind load. The weather-aware model is trained and served, and answers as soon as weather catches up - it is not dead code, it is starved.

3. Are the accuracy figures real?

A published metric can be wrong in two ways: computed incorrectly, or computed on data the model had already seen. scripts/verify_model_accuracy.py recomputes every accuracy figure on this page from training_features.parquet and the committed classifier without reading closed_loop_report.json. Its gates are property-based - the integrity sections must hold exactly and the later ones fail only when two independent computations of the same quantity disagree - so a threshold cannot be moved to make a result look better. This build ran its offline subset:

```
python scripts/verify_model_accuracy.py --skip-live
checks run 10 | passed 10 | failed 0
```

The label matches its documented rule exactly, no feature can read the label off the frame, the scored window strictly postdates the fitted window, and the held-out metrics above were recomputed independently of the served report. The live API comparison is not in this count - run the command without --skip-live for that.

- The shuffle-before-split recipe some reviewers expect was measured, not argued: random shuffle puts 10,392 of 10,392 scored rows (100.0%) chronologically at or before the last fitted hour, and 41,349 of the 41,567 fitted rows (99.5%) take their lag/rolling inputs directly from held-out rows. A temporal split is therefore the only defensible choice for this data, and it is the one the trainer uses.
- The aggregate recall is a blend that hides a zone: Italy South scores precision 0.8130 / recall 0.4556 over 3,374 held-out hours (439 real overloads). Read per-zone numbers, not one percentage.
- Extreme-overload blindness (ES ladder): no rung reaches the 0.6 alarm line; the highest score in the ladder is 0.490872 at util 1.30. A tree ensemble cannot emit a leaf value it never trained on, so an overload worse than anything in the two recorded years is NOT flagged.

4. What was added

Backend - 26 commits (5 of them capability changes):

- f78b76e feat(api): refresh the served frame at request time instead of serving a frozen commit
- f91fd2f feat(reports): emit the submission artifacts from live measurements
- 0128745 feat(api): serve the fallback profile and declare one horizon per feed in /health
- 682615c feat(models): retrain on the real window and add a calibrated no-weather fallback
- 9c6a48e feat(data): rebuild the feature frame on two years of real ENTSO-E and NASA POWER hours

Plus 15 documentation, test or build commits in the same window.

Frontend - 17 commits (9 of them capability changes):

- 37b11b0 feat(mobile): replace scroll-to-section nav with five real tab views
- e2c3d38 feat(ui): measured colophon footer with provenance and rights
- 2615d94 feat(ui): show the scale the risk score is measured against
- 14b1dfb feat(ui): say what the product does, next to the wordmark
- a7a9284 feat(ui): give the phone first screen something worth reading
- ea21c69 feat(ui): a phone layout that reorders the dashboard instead of shrinking it
- b0b0190 feat(3d): radar sweep with radial falloff, selection reticle and honest frame timing
- 8a53e54 feat(ui): one stroked SVG icon family, loading skeletons and real favicons
- 3b73f80 feat(ui): signal integrity rail stating what each feed actually reaches

Plus 2 documentation, test or build commits in the same window.

5. What broke, and what fixed it

12 commits in the window are fixes. Each one below was found by running the system and measuring the result, not by reading the code.

- 52edea9 fix(refresh): stop shipping a timeout below the endpoint's own response floor - The first deployed run of the request-time refresher reached ENTSO-E with the
- b8887a9 fix(security): redact the ingestion script's own failure print too - The rule "every message that quotes an exception is scrubbed before it is
- 2eea94f fix(refresh): make one real HTTP attempt, and never publish the token - Two defects in the request-time refresh, both found by running the server
- fd35c7e fix(api): bound the reflected zone echo and set security + no-store headers - Probing the deployed API on 2026-11-06 showed two defects that reading the code
- fcf0744 fix(validation): publish repo-relative artifact paths on a public endpoint - /validation echoed str(MODELS_DIR / name), which exposed the absolute
- 43fa10d fix(models): refuse to forecast weather-needing models from a weather-less frame - The no-weather path fills absent weather columns with zeros, which is only sound while

- 8e01149 fix(copy): stop claiming liveness the serving path does not have - The API loads the committed feature frame at startup and makes no
- b44f979 fix(3d): stop warning about the shadow type three already substitutes - three 0.185 logs PCFSoftShadowMap as deprecated and silently uses PCFShadowMap, so selecting PCF explicitly changes nothing in the render but clears t
- 028aa84 fix(ui): show release gates as a map, not a 3px hairline - The summary bar was measured at 3 CSS pixels tall, which reads as a divider rather than a result, so 9 PASS / 4 WARN looked like missing content. One
- e524e3c fix(chart): restore the phone forecast plot and the median line - The 24h plot rendered as a ~470px void below lg. The chart box was a
- 8b77cbf fix(copy): say what each feed is, not just "proxy" - Bare "PROXY" read as an apology for the data. The load and weather feeds
- 2deb39c fix(ui): wrap instead of clip, hit 32px targets, and label what colour alone cannot - Measured against the real feed at 375, 768 and 1440 px, the panel pass removed the

Two of the measured defects are worth calling out because a screenshot would not have caught either of them:

- The phone forecast panel was a ~470px void. A flex item sized by min-height clamping does not have a definite height, so the chart container's height:100% resolved to 0. Measured: container 331x0, no SVG surface. Now 331x280 with a drawn surface.
- The P50 median line had never rendered at any width. Recharts' AreaChart renders only Area children and silently drops Line children - no warning, no error, no console output. Switched to ComposedChart; measured 3 line paths, median stroke #35d6e8, dasharray none.

6. What remains open

The release gates that are WARN, not PASS (4 of 13):

- weekly coverage spread <=0.15 (band not decaying across the window)
- >=80% of real overload events raise an alarm while running
- >=50% of events forecast-warned >=3h before onset
- median STABLE forecast lead before onset >=3h (unbroken warning from h+1)

Beyond the gates, 4 things are genuinely unfinished:

- Data currency, which is now a credential question rather than a missing stage. There is a refresh path: the API answers from the committed frame and a request-time refresher runs on top of it, at most one pass per 900 s window, per zone only when its newest served hour is older than 3.0 h, under a 30 s ceiling, and falling back to the committed frame on any failure. Live at build time it measures running, 3/3 zones answered 401 Unauthorized - 0 rows published, last attempt 2026-11-06T12:36:20.932910+00:00 taking 30.6 s, with last success null. The newest served hour is therefore still 2026-11-04T17:00:00+00:00, 43.7 h old; the frame shipped at 9c6a48e was already 25.9 h old. /health and the dashboard report these ages truthfully, so nothing here is mislabelled as current.
- Rated capacity. Every MWa figure is a p99-of-load proxy derived from the same load series it scores, so utilisation percentages are self-referential. This is stated in the UI, the model card and the audit report; it is not fixed.
- Weather currency. NASA POWER arrives on a multi-day lag, so the deployed product runs on the fallback model. The weather-aware path is therefore exercised in replay, not in production serving.
- Domain. Load is ENTSO-E (Spain, Greece, Italy South) and weather is NASA POWER. There is no public Pakistani hourly feed to train on, so NTDC/DISCO is stood in for and no accuracy figure here transfers to it without retraining.

Closed since the previous note: the numbered docs are synced. None of the 6 superseded-figure patterns (synthetic-training claims, 13/13 PASS, the 206-event count, the 3.0 h lead, the 48-test count, and the claim that the runtime makes no outbound fetch or holds no credential) match anywhere in docs/, checked by grep when this page was built. The limit of that check is its pattern list - prose can still drift in ways none of these expressions catch, so this is evidence the known false claims are gone, not proof the docs are exhaustively correct.

7. Future direction

Ordered by what each step would actually change about the numbers on screen.

- Rated capacity per zone first. It is the cheapest high-impact change: it converts utilisation and risk from a self-referential scale into one an operator can act on. Needs a published transmission capacity figure per bidding zone, or TSO network development plan data.
- Make one refresh pass publish. The blocker is not code: 3/3 zones are answered 401 Unauthorized by the ENTSO-E RESTful API, so the secret in the deployment platform's store must be replaced with a token whose RESTful-API access ENTSO-E has approved. Until then every cache window spends one 30 s attempt on a credential that cannot succeed, which

GRIDSENSE_REFRESH=false stops. Shipping the refresher bought a fresher frame and cost two properties the committed parquet had: the runtime now holds a third-party credential and does make an outbound request at request time. Both are bounded - GET-only serving, one attempt per zone per window, the timeout ceiling, a redacted last_error, and the committed-frame fallback.

- A forward-looking weather feed with a horizon that matches the forecast horizon. One candidate was evaluated and rejected on a measured variable mismatch against the trained feature set; adopting any such feed means retraining, and the closed-loop replay is already in place to show whether the swap helps.
- Close the two WARN gates rather than accept them, then re-run the negative control so a passing board still means something.
- Re-point the same pipeline at a non-European zone to prove it generalises: the feature builder, model set and replay engine are zone-parameterised, so the second domain is a data task, not a rewrite.
- Only then consider real-time alerts or a dispatch-facing surface. Nothing in the current evidence supports turning a research prototype into an operational instruction, and the UI says so on every page.

8. How to re-check this note

```
curl -s https://grid-senseai-backend.fastapicloud.dev/health | python -m json.tool | head -40
curl -s https://grid-senseai-backend.fastapicloud.dev/validation | python -c "import json,sys; d=json.load(sys.stdin); print(d['overall'], [g['result'] for g in d['gates']])"
python -c "import pandas as pd; f=pd.read_parquet('data/features/training_features.parquet'); print(len(f), f['zone'].nunique())"   # trained rows / zones
python scripts/verify_model_accuracy.py          # 22 checks, incl. the live API
git -C gridsense-backend log --since='2026-11-05 00:00' --oneline
git -C gridsense-frontend log --since='2026-11-05 00:00' --oneline
python scripts/build_release_note_pdf.py   # rebuilds this file
```

Sources: backend https://github.com/huzaifahimad/Grid_Sense.AI-Backend- ; frontend https://github.com/huzaifahimad/Grid_Sense.AI-frontend- ; built by build_release_note_pdf.py on 2026-11-06 12:44 UTC.